

Lightning Solvers in Sloshing Modes

Pehan Amarathunge

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1 Introduction

The sloshing problem is concerned with the motion of the free surface of a liquid inside a rigid container due to some kind of disturbance. The basic problem of liquid sloshing involves determining the natural frequencies of oscillation and the modes of an impermeable, rigid container.

Sloshing is of paramount importance in engineering. For example, the sloshing of propellant in the fuel tanks in vehicles, aircrafts and spacecrafts could have a significant impact on critical maneuvers such as the docking of cargo ships or the grounding of spacecrafts. For example, sloshing of the remaining propellant induced an oscillatory motion of the Apollo-11 Lunar Module in 1969, which hampered control during landing [1]. Knowledge of the sloshing mode frequencies would be highly beneficial for space missions, as the control systems frequencies must be carefully selected not to excite the sloshing modes [2]. Furthermore, sloshing must be accounted for in the seismic design of liquid storage tanks, and severe sloshing may lead to the liquid leakage or buckling of the steel tank wall. For instance, the 2003 Tokachi-oki earthquake in Japan damaged a large tank storing oil at an industrial facility, causing a fire [3]. For further applications, the reader is directed to books by Faltinsen and Timokha (Chapter 1) [4] as well as the book by Ibrahim [5].

In this paper, we are concerned with numerically determining the sloshing modes and frequencies of a two-dimensional tank of arbitrary shape. Furthermore, the fluid is assumed to be inviscid, incompressible and irrotational. The fluid is filled in a rigid, impermeable container and makes small oscillations about the equilibrium position. The equations of motions can be linearized and the velocity potential can be expressed as $\Re(\phi \exp(-i\omega t))$. The cartesian axes are chosen such that the y-axis lies vertically upwards. Then the governing equation for ϕ inside the container will be Laplace's Equation:

$$\nabla^2 \phi = 0. \tag{1}$$

The boundary condition on the rigid, impermeable walls is:

$$\frac{\partial \phi}{\partial n} = 0. \tag{2}$$

The boundary condition on the free surface is:

$$\frac{\partial \phi}{\partial y} = \lambda \phi, \quad (3)$$

where the eigenvalue λ is related to the frequencies of oscillation, denoted as ω by $\omega^2 = \lambda g$. A full derivation of the equation and boundary conditions can be found in [4] or [5].

Despite the significance of this problem, there are only a handful of container shapes whose sloshing frequencies can be determined analytically in a closed-form expression. Notable examples, include the separable solution of the rectangle [6] and Kirchoff’s solution for a triangle whose sides subtend an angle of 45 degrees from the free surface [7]. We also note that solutions also exist for the even modes of an isosceles triangle whose sides subtend an angle of 30 degrees [8]. Furthermore, Anthony Davis [9] showed that in the case of the triangle, there are no other closed form solutions for all other angles that are a rational multiple of π .

As a result, we will have to resort to numerical methods, but nevertheless, its numerical solution is challenging in many domains of practical interest. Typically, numerical methods struggle when the boundary of the domain is not smooth and admits a corner singularity, which hinders traditional techniques such as finite-element methods and boundary element methods. Although these approaches have been adapted to account for corners [10], their implementation is complex and requires specialist knowledge.

Recently, a new method proposed by Trefethen [11] uses rational function approximation theory to solve Laplace’s equation in domains with corners. This technique is extremely fast, accurate and straightforward to implement. Its applications to a spectrum of potential flow problems in fluid mechanics were demonstrated by Peter Baddoo [12]. In this paper, we will examine a particular application of the lightning solver to sloshing problems. The principal advantages of our methods revolve around its simplicity and generality. It is simple in the sense that it does not require specialist knowledge to implement in domains with corners, as opposed to other typical numerical methods that have been outlined above. Secondly, it is general, as it can be applied to determine the sloshing modes and frequencies of an arbitrarily shaped two-dimensional rigid container.

References

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